ARNAB GHOSH

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LinkedIn | GitHub | HackerRank | LeetCode | Portfolio

SUMMARY

Final-year B.Tech student in Computer Science and Business Systems with hands-on experience in Machine Learning, Deep Learning, and Natural Language Processing (NLP). Built end-to-end projects involving emotion classification using Bi-GRU and LSTM neural networks, mental health score prediction using Random Forest, and NLP-based movie recommendations using TF-IDF and cosine similarity. Proficient in Python, Pandas, NumPy, Scikit-learn, TensorFlow/Keras, and FastAPI, with experience in model development, data preprocessing, evaluation, and deployment.

TECHNICAL SKILLS

Languages & Core CS: Python, SQL (MySQL), Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP)

Data Analysis & Processing: Pandas, NumPy, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering

Machine Learning & Deep Learning: Scikit-learn, XGBoost, Regression, Classification, Clustering, Random Forest, Model Evaluation, Hyperparameter Tuning, TensorFlow/Keras, LSTM, Bi-GRU

NLP & Recommendation Systems: Natural Language Processing (NLP), Text Preprocessing, Tokenization, TF-IDF, Cosine Similarity, Text Classification

Tools & Deployment: FastAPI, Streamlit, Jupyter Notebook, Google Colab, Git, GitHub, Render

PROJECTS

Emotion Prediction – Deep Learning NLP Application **[Link]**

Python, NLP, TensorFlow/Keras, Bi-GRU, LSTM, FastAPI, HTML, CSS, JavaScript, Render

- Developed a **deep learning application** to classify text into different emotion categories using NLP techniques.
- Implemented **Bi-GRU and LSTM** models using TensorFlow/Keras to learn contextual patterns in text.
- Built a text preprocessing pipeline with tokenization and sequence padding, and developed a **FastAPI** backend for real-time predictions.
- Deployed the application on **Render** with a responsive web interface for interactive emotion prediction.

Mental Health Score Predictor – Machine Learning **[Link]**

Python, Pandas, NumPy, Scikit-learn, Random Forest, RandomizedSearchCV, FastAPI, Pydantic, JavaScript, Render

- Developed an **end-to-end ML application** to predict mental health scores from lifestyle and behavioral data.
- Performed preprocessing, outlier handling, categorical encoding, and log transformation to prepare features.
- Trained a **Random Forest Regressor** and tuned it with **RandomizedSearchCV** for better performance.
- Built a REST API using **FastAPI and Pydantic** and deployed an interactive web application on **Render**.

EDUCATION

B.Tech in Computer Science and Business Systems | Academy of Technology, Kolkata — Sep 2023 - Jun 2027

- SGPA: 7.9 (through 6th semester)

Higher Secondary Examination | Bohar High School (WBCHSE) — Apr 2021 - May 2022

- Grade: A+

CERTIFICATIONS

Python Certification — [Link]

Machine Learning Certification — [Link]

Power BI Certification — [Link]

HackerRank SQL Certification — [Link]

ACHIEVEMENTS

- 2nd Runner-Up at **TechSprint**, organized by GDGoC (Google Developer Groups on Campus), Academy of Technology.
- Shortlisted at the campus level for **Smart India Hackathon (SIH) 2025**.

LANGUAGES

English (Fluent), Bengali (Native), Hindi (Intermediate)